
JAL240I

PRINCIPLES OF ARTIFICIAL INTELLIGENCE

UNIT V

EXPERT SYSTEMS AND APPLICATIONS OF ARTIFICIAL INTELLIGENCE

UNIT V - EXPERT SYSTEMS AND APPLICATIONS OF AI

SYLLABUS

Expert systems - Architecture of expert systems - Roles of expert systems - MYCIN - DART - XOON - Application Domains Of AI - Artificial Intelligence in Environmental Management - Artificial Intelligence in Systems Biology: Opportunities in Agriculture, Biomedicine and Healthcare - Artificial Intelligence in Disease Diagnosis via Smartphone Applications.

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EXPERT SYSTEMS



EXPERT SYSTEMS

- **Definition:** An expert system is AI software that uses knowledge stored in a knowledge base to solve problems that would usually require a human expert thus preserving a human expert's knowledge in its knowledge base.
- **Definition:** The expert systems are the computer applications developed to solve complex problems in a particular domain, at the level of extraordinary human intelligence and expertise.
- They can advise users as well as provide explanations to them about how they reached a particular conclusion or advice.
- **Knowledge Engineering** is the term used to define the process of building an Expert System and its practitioners are called **Knowledge Engineers**.

EXPERT SYSTEMS

- The primary role of a knowledge engineer is to make sure that the **computer possesses all the knowledge** required to solve a problem.
- The knowledge engineer must choose one or more forms in which to represent the required knowledge as a **symbolic pattern** in the memory of the computer.
- A good Expert System solves a problem accurately, quickly and is easy to use.
- Often used to help non-experts when a human expert is too expensive, the results too slow if use a human(s), error rate too high with a human(s), unintentional human bias, or it is difficult for a person to reach the location.

EXPERT SYSTEMS - EXAMPLES

MYCIN

- One of the earliest expert systems based on backward chaining. It can identify various bacteria that can cause severe infections and can also recommend drugs based on the symptoms.

DENDRAL

- It was an artificial intelligence-based expert system used for chemical analysis. It used a substance's spectrographic data to predict its molecular structure.

RI/XCON

- It could select specific software to generate a computer system configuration wished by the user.

PXDES

- It could easily determine the type and the degree of lung cancer in a patient based on the data.

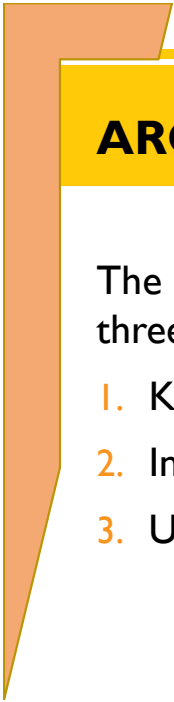
CaDet

- It is a clinical support system that could identify cancer in its early stages in patients.

Dxplain

- It was also a clinical support system that could suggest a variety of diseases based on the findings of the doctor.

ARCHITECTURE OF EXPERT SYSTEMS

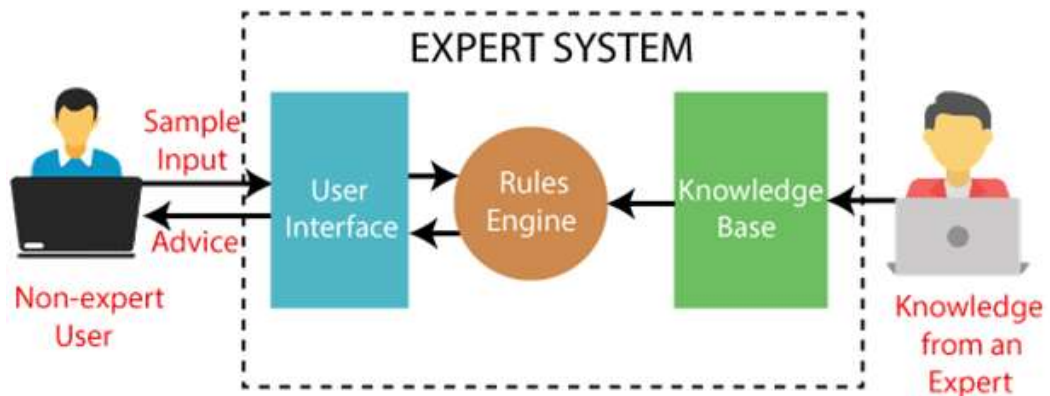


ARCHITECTURE OF EXPERT SYSTEMS

The architecture of expert system in AI is typically consists of three main components:

1. Knowledge Base
2. Inference Engine
3. User Interface

ARCHITECTURE OF EXPERT SYSTEMS



ARCHITECTURE OF EXPERT SYSTEMS

- **Knowledge Base:** This component stores the domain-specific information, rules, and facts that the expert system uses to make decisions and solve problems. The knowledge base is created by human experts in the domain and is a critical part of the system. It contains both declarative knowledge (facts about the domain) and procedural knowledge (rules and heuristics for problem-solving).
- **Inference Engine:** The inference engine is responsible for processing the information in the knowledge base and applying it to specific situations or problems. It uses various reasoning techniques, such as forward chaining and backward chaining, to derive conclusions and make decisions. The inference engine evaluates the available data and rules to provide recommendations or solutions.

ARCHITECTURE OF EXPERT SYSTEMS

- **User Interface:** The user interface component allows users or operators to interact with the expert system. It can take the form of a text-based interface or a graphical user interface (GUI). Users can input questions or problem descriptions, and the expert system uses its knowledge base and inference engine to provide answers, solutions, or advice.
- **Explanation Module:** This module helps the expert system to give the user an explanation about how the expert system reached a particular conclusion. The explanation module enhances transparency in decision-making. It provides users with explanations for the system's recommendations or decisions. This is particularly important because it allows users to understand why the system arrived at a specific outcome, increasing their confidence in the expert system's suggestions.

ARCHITECTURE OF EXPERT SYSTEMS

- Expert systems are built upon a knowledge base, which stores domain-specific information, facts, and rules. It is the repository of facts, rules, and domain-specific information. This knowledge base represents the collective wisdom and expertise of human experts in the chosen domain.
- The inference engine, simulates the human decision-making process. It uses the knowledge stored in the knowledge base to draw inferences, make decisions, and provide solutions.
- Expert systems replicate human problem-solving by employing reasoning mechanisms like deductive, inductive, or abductive reasoning.

ROLES OF EXPERT SYSTEMS

ROLES OF EXPERT SYSTEMS

- ❖ Advising
- ❖ Instructing and assisting human in decision making
- ❖ Demonstrating
- ❖ Deriving a solution
- ❖ Diagnosing
- ❖ Explaining
- ❖ Interpreting input
- ❖ Predicting results
- ❖ Justifying the conclusion
- ❖ Suggesting alternative options to a problem

EXPERT SYSTEMS

MYCIN - DART - XOON

MYCIN

- MYCIN, an early expert system, or AI program, for treating blood infections.
- In 1972 work began on MYCIN at Stanford University in California.
- MYCIN would attempt to diagnose patients based on reported symptoms and medical test results.
- The program could request further information concerning the patient, as well as suggest additional laboratory tests, to arrive at a probable diagnosis, after which it would recommend a course of treatment.
- If requested, MYCIN would explain the reasoning that led to its diagnosis and recommendation.
- Using about 500 production rules, MYCIN operated at roughly the same level of competence as human specialists in blood infections and rather better than general practitioners.

MYCIN

- MYCIN was an early backward chaining expert system that used artificial intelligence to identify bacteria causing severe infections, such as bacteremia and meningitis, and to recommend antibiotics, with the dosage adjusted for patient's body weight — the name derived from the antibiotics themselves, as many antibiotics have the suffix "-mycin".
- The Mycin system was also used for the diagnosis of blood clotting diseases. MYCIN was developed over five or six years in the early 1970s at Stanford University.
- It was written in Lisp as the doctoral dissertation of Edward Shortliffe under the direction of Bruce G. Buchanan, Stanley N. Cohen and others.
- It arose in the laboratory that had created the earlier Dendral expert system.
- MYCIN was never actually used in practice but research indicated that it proposed an acceptable therapy in about 69% of cases, which was better than the performance of infectious disease experts who were judged using the same criteria.

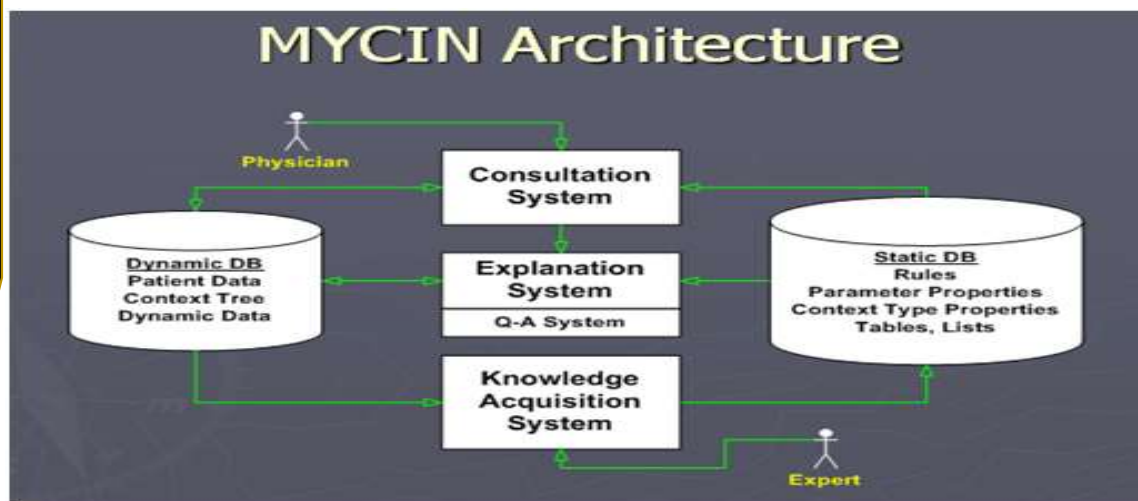
MYCIN

- MYCIN operated using fairly simple inference engine, and a knowledge base rules.
- It would query the physician running the program via a long series of simple yes/no or textual questions.
- At the end, it provided a list of possible culprit bacteria ranked from high to low based on the probability of each diagnosis, its confidence in each diagnosis' probability,
- The reasoning behind each diagnosis (that is, MYCIN would also list the questions and rules which led it to rank a diagnosis a particular way), and its recommended course of drug treatment.

MYCIN

- Despite MYCIN's success, it sparked debate about the use of its ad hoc, but principled, uncertainty framework known as "certainty factors".
- The developers performed studies showing that MYCIN's performance was minimally affected by perturbations in the uncertainty metrics associated with individual rules, suggesting that the power in the system was related more to its knowledge representation and reasoning scheme than to the details of its numerical uncertainty model.
- Some observers felt that it should have been possible to use classical Bayesian statistics. MYCIN's developers argued that this would require either unrealistic assumptions of probabilistic independence, or require the experts to provide estimates for an unfeasibly large number of conditional probabilities.

MYCIN



MYCIN - CONSULTATION SYSTEM

Performs Diagnosis and Therapy Selection	User-Friendly Features	Control Structure
Control Structure reads Static DB (rules) and read/writes to Dynamic DB (patient, context)	Users can request rephrasing of questions	Goal-directed Backward-chaining Depth-first Tree Search
Linked to Explanations	Synonym dictionary allows latitude of user responses	High-level Algorithm: • Determine if Patient has significant infection • Determine likely identity of significant organisms • Decide which drugs are potentially useful • Select best drug or coverage of drugs
Terminal interface to Physician	User typos are automatically fixed	
	Questions are asked when more data is needed	
	If data cannot be provided, system ignores relevant rules	

MYCIN - STATIC DATABASE

- Rules
- Meta-Rules
- Templates
- Rule Properties
- Context Properties
- Fed from Knowledge Acquisition System

MYCIN - PRODUCTION RULES

- Represent Domain-specific Knowledge
- Over 450 rules in MYCIN
- Premise-Action (If-Then) Form:
 <predicate function> <object> <attrib> <value>
- Each rule is completely modular, all relevant context is contained in the rule with explicitly stated premises

Assumptions

- Not every domain can be represented, requires formalization (EMYCIN)
- Only small number of simultaneous factors (more than 6 was thought to be unwieldy)
- IF-THEN formalism is suitable for Expert Knowledge Acquisition and Explanation sub-systems

DART

- Dynamic Analysis and Replanning Tool
- DART is a joint project of the Heuristic Programming Project and IBM that explores the application of artificial intelligence techniques to the diagnosis of computer faults.
- The primary goal of the DART Project is to develop programs that capture the special design knowledge and diagnostic abilities of these experts and to make them available to field engineers.
- The practical goal is the construction of an automated diagnostician capable of pinpointing the functional units responsible for observed malfunctions in arbitrary system configurations.

DART



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DART

- DART uses intelligent agents to aid decision support system • Give planners the ability to rapidly evaluate plans for logistical feasibility.
- DART decreases the cost and time required to implement decisions.
- The field engineer is familiar with the diagnostic equipment and software testing.
- Access to information about the specific system hardware and software configuration of the installation.

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XCON

- The RI (internally called XCON, for eXpert CONfigurer) program was a production rule based system written in OPS5 by John P. McDermott of CMU in 1978.
- Configuration of DEC VAX computer systems
- Ordering of DEC's VAX computer systems by automatically selecting the computer system components based on the customer's requirements.
- XCON first went into use in 1980 in DEC's plant in Salem, New Hampshire. It eventually had about 2500 rules.
- By 1986, it had processed 80,000 orders, and achieved 95.98% accuracy.
- It was estimated to be saving DEC \$25M a year by reducing the need to give customers free components when technicians made errors, by speeding the assembly process, and by increasing customer satisfaction.

XCON

- XCON interacted with the sales person, asking critical questions before printing out a coherent and workable system specification/order slip.
- XCON's success led DEC to rewrite XCON as XSELa version of XCON intended for use by DEC's salesforce to aid a customer in properly configuring their VAX.

APPLICATION DOMAINS OF AI

APPLICATION DOMAINS OF AI

Healthcare

- Disease diagnosis, drug discovery, personalized treatment, virtual health assistants

Finance

- Fraud detection, algorithmic trading, credit risk assessment, customer service automation

Retail & E-Commerce

- Chatbots, personalized recommendations, inventory management, price optimization

Transportation

- Self-driving cars, traffic prediction, logistics optimization, smart public transport

Manufacturing

- Predictive maintenance, robotics, supply chain optimization

APPLICATION DOMAINS OF AI

Cybersecurity

- Threat detection, AI-driven fraud prevention, secure authentication

Education

- AI tutors, personalized learning, automated grading, smart content generation

Agriculture

- Precision farming, crop health monitoring, automated harvesting

Environmental Management

- Climate change monitoring, disaster prediction, wildlife conservation

Entertainment & Media

- AI-generated content, personalized streaming recommendations, deepfake detection

ARTIFICIAL INTELLIGENCE IN ENVIRONMENTAL MANAGEMENT

AI IN ENVIRONMENTAL MANAGEMENT

- AI is being used in every field.
- AI algorithm guides us to do our day-to-day task.
- Focus of AI in environmental management is to give a flavor of what AI could do in different areas, directly impacting environment.
- A proactive approach deals in solving for the root cause of the problem → AI used to control the climate.
- Reactive approach (or) risk mitigation approach deals in solving the occurred problem such as predicting extreme weathers and disasters to prepare to deal with these problems.
- Air pollution is a silent killer and is the reason for a wide range of health diseases such as respiratory difficulties, heart malfunctions and strokes → By reducing air pollution levels, countries can reduce stroke, heart disease, lung cancer, and both chronic and acute respiratory diseases including asthma and untimely deaths.

AI IN ENVIRONMENTAL MANAGEMENT

Various applications of AI in environmental management

- AI for Cleaner Air – Smart Pollution Control
- AI for Water Preservation – Smart Water Management
- AI for Better Agriculture – Smart Farming
- AI for Better e-Waste Management – Smart Monitoring/Control
- AI for Climate Control – Smart Energy Optimization

AI IN ENVIRONMENTAL MANAGEMENT

Risks and Rewards of AI in Environmental Management

- In recent years, **AI technologies** have played a significant role in addressing **climate change**. Applications include the following:
 - **Greenhouse Gas (GHG) Reduction:** AI supports forecasting and control in sectors like electricity, transportation, and industry to lower emissions.
 - **Smart Energy Systems:** AI uses historical and real-time data to predict power generation and optimize energy usage.
 - **Sustainable Transport:** Helps improve vehicle efficiency and shift to low-carbon transport options.

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AI IN ENVIRONMENTAL MANAGEMENT

- **Smart Cities and Buildings:** Assists in designing energy-efficient infrastructure and managing heating/cooling systems.
- **Industrial Optimization:** Enhances production quality, prevents breakdowns, and supports clean energy transitions.
- **Precision Agriculture:** Reduces emissions from soil, boosts yield, predicts disasters like forest fires, and supports carbon absorption.
- **Climate Forecasting:** Enables better planning by predicting climatic changes and encouraging the use of clean energy.
- The effectiveness and responsibility of AI systems lie in the hands of humans who design, integrate, and govern them. Responsible AI ensures that the risks are minimized while maximizing societal and environmental rewards.

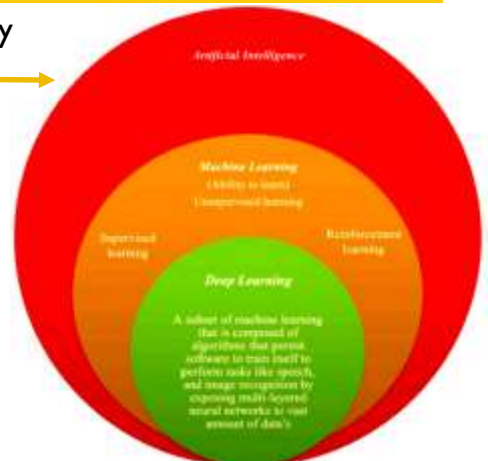
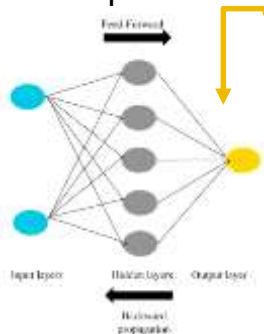
ARTIFICIAL INTELLIGENCE IN SYSTEMS BIOLOGY: OPPORTUNITIES IN AGRICULTURE, BIOMEDICINE AND HEALTHCARE

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AI IN SYSTEMS BIOLOGY: OPPORTUNITIES IN AGRICULTURE, BIOMEDICINE AND HEALTHCARE

AI Methodologies and Algorithm for Systems Biology

- Machine Learning (ML)
- ML Algorithm Applied in Systems Biology
- Computational Neural Networks



AI IN SYSTEMS BIOLOGY: OPPORTUNITIES IN AGRICULTURE, BIOMEDICINE AND HEALTHCARE

Sub-disciplines of AI



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AI IN SYSTEMS BIOLOGY: OPPORTUNITIES IN AGRICULTURE, BIOMEDICINE AND HEALTHCARE

Applications of Artificial Intelligence (AI) in Agriculture, Biomedicine, and Healthcare

- AI in Agriculture
- AI in Biomedicine
- AI in DNA Expression Profiling
- AI for Identifying Exonic Regions
- AI in Identifying Variants/Mutations from Genetic Data
- AI Workflow Method for Genomic Analysis
- AI in Structure Prediction
- AI in Phylogeny
- AI in Healthcare
- Case study 1 → AI in PHARMACOGENOMICS
- Case study 2 → AI in Systems Biology for Cancer Cure
- Case study 3 → Applications of AI for COVID-19 Pandemic

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ARTIFICIAL INTELLIGENCE IN DISEASE DIAGNOSIS VIA SMARTPHONE APPLICATIONS



AI IN DISEASE DIAGNOSIS VIA SMARTPHONE APPLICATIONS

Smartphone Applications and ML Algorithms in Disease Diagnosis

- Diagnosis of Diseases by Using Smartphone Applications
- Smartphone App for Noninvasive Detection of Anemia
- Mobile Touch Screen Typing in the Detection of Motor Impairment of Parkinson's Disease
- Screening Services for Cancer on Android Smartphones
- Detection of Cardiovascular Disease Using Smartphone Mechanocardiography
- Mobile-Enabled Expert System for Diagnosis of Tuberculosis in Real Time
- Smartphone-Based Pathogen Detection of Urinary Sepsis
- Detecting Acute Otitis Media Using a Mobile App
- Diagnosis of Covert Hepatic Encephalopathy via Encephal App, a Smartphone-Based Stroop Test
- Mobile-Based Nutrition and Child Health Monitoring

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AI IN DISEASE DIAGNOSIS VIA SMARTPHONE APPLICATIONS

Machine Learning Technology in the Diagnosis of Various Diseases

- Heart Disease
- Diabetes Disease
- Liver Disease
- Dengue Disease
- Hepatitis Disease
- Genetic Disorders (Acromegaly) from Facial Photographs.

